Performance Testing

Date	23 October 2026
Team ID	[Your Team ID]
Project Name	TravelMate
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	**Details**
Testing Tool Used	Postman and Locust (planned); browser Lighthouse for frontend load time
Type of Testing	Load Testing
Target Module / API	Backend /health endpoint, trips API and AI itinerary generation endpoint
Test Environment	Hosted: Render backend (https://personalized-travel-itinerary-planner-k0ey.onrender.com) and Vercel frontend
Test Date	To be filled after the test run

Step 2: Test Scenarios

S.No	**Test Scenario / Description**	**No. of Virtual Users**	**Duration (sec)**	**Expected Outcome**
1	Health check: repeated GET /health requests	10	60	Every response returns status ok.
2	Authenticated user loads the trips list	10	60	Trips list returns within target response time.
3	Generate an AI itinerary for a sample trip	3	120	Valid structured itinerary returned without errors.
4	Open the frontend homepage and dashboard	5	60	Pages render within target load time.

Step 3: Performance Test Results

S.No	**Metric**	**Target Value**	**Actual Value**	**Status (Pass / Fail)**	**Remarks**
1	Response Time (Avg)	< 2 seconds	Pending	Pending	Excludes AI generation

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
2	Response Time (Max)	< 5 seconds	Pending	Pending	
3	Throughput (Req/sec)	Baseline to be set	Pending	Pending	
4	Error Rate	< 1%	Pending	Pending	
5	CPU Utilization	< 80%	Pending	Pending	From Render metrics
6	Memory Utilization	< 80%	Pending	Pending	From Render metrics

Step 4: Observations & Analysis

The tests have not been executed yet. Results will be recorded after production verification. Factors to watch: AI generation time depends on the Groq API, geocoding depends on Nominatim rate limits (a coordinate cache and rate limiting are already implemented), and the hosted backend may respond slowly on its first request.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., Locust graphs, Postman runner, Lighthouse report):

[Insert screenshot]

Add performance testing result screenshots here